

Linux Programming: Assignment-1: 20-09-2025

1. What is Linux Operating System (OS)? List three pros and cons of it. (CO1)
2. Differentiate between **Linux, Mac, Android,** and **Windows OS** with at least **six** unique features. (CO1)
3. Why is Linux preferred for **Mainframe Servers** for legacy application? Give three out-of-the-box technical reasons. (CO1)
4. Explain the structure of the Linux File System with proper diagram. Note: you can use the **tree** command to find it out. (CO2)
5. If Linux OS is open-source, how do companies like Red Hat still making money from it? Do a market study and answer properly. (CO2)
6. Write the **command** to display today's date and time (i.e., current System time). (CO1)
7. Which **command** is used to check how long the system has been running? (CO1)
8. What is the difference between **shutdown -h** now and **halt**? (CO1)
9. Compare **init 0** and **shutdown -h**. Which is safer? Why? (CO1)
10. A system administrator accidentally powers off a Server machine without shutting it down properly. What problems can occur to the said Server? (CO2)

Brainstorming

- a) As Linux Kernel is open-source, can we build our own operating system?
- b) In order to do that, what are the stoppers, hurdles, and challenges?
- c) Is anyone in India working on this field? Find at-least three to four engineers.

Note:

- (i) All the above questions are from Module-1. Refer your textbook or reference book to answer all. You can use Open AI Tools also. But answer must be decorated in your own language.
- (ii) We will accept softcopy from your end but it will go through content similarity checking through TURNITIN Software. So, don't try to copy paste from the Internet. If we find similarity your internal marks will be less.
- (iii) Each question carries 3 Marks: Total 30 Marks. **Submission Deadline: 23rd September, 2025.**